REVIEW PAPER

Department of Artificial Intelligence

Encrypted Traffic Classification

A Comparative Review of ATVITSC, AutoML4ETC and MLETC

Deep learning approaches for classifying encrypted network traffic

SUBMITTED BY

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Abstract

Encrypted traffic now dominates the Internet, making port-based and payload-based classification largely ineffective and pushing researchers towards machine learning and deep learning. This paper reviews three recent studies that approach the problem differently. ATVITSC converts sessions into packet-level images and fuses a vision transformer with a convolutional and bidirectional LSTM branch. AutoML4ETC uses neural architecture search to generate compact classifiers for early classification of TLS and QUIC flows. MLETC pre-trains a DeBERTa model on byte, field and packet levels of traffic. The studies are compared by problem, method, datasets, results, strengths and weaknesses. ATVITSC and MLETC report roughly 98–99% accuracy on public benchmarks, while AutoML4ETC reports 82–96% service-level accuracy on harder operator traffic with far smaller models. The review concludes that benchmark accuracy is nearly saturated and that progress now depends on lightweight models, privacy-preserving training, cross-domain generalisation, explainability and real-world evaluation.

Keywords: Encrypted Traffic Classification, Network Security, Deep Learning, Machine Learning

1. Introduction

Traffic classification assigns network flows or packets to a category, such as an application (Skype, Netflix), a service type (chat, streaming, file transfer), a tunnelling technology (VPN, Tor) or a security class (benign or malicious). It supports resource planning, quality-of-service control, intrusion detection and malware detection [1], [2].

Approaches fall into four generations. Port-based methods map well-known ports to applications but fail with dynamic ports. Payload-based methods, known as Deep Packet Inspection (DPI), search packet contents but cannot read encrypted payloads and are costly. Classical machine learning, for example random forests, uses hand-designed statistical features such as packet size and timing, so its quality depends on feature design. Deep learning learns features directly from raw bytes, images or sequences [1], [3]. Within deep learning, the field moved from CNNs and RNNs, such as the 1D-CNN [8], FS-Net [6] and Deep Packet [5], to hybrid CNN–LSTM models [7], pre-trained transformers such as ET-BERT [4], and recently automated architecture design. This review examines three papers that reflect these trends: ATVITSC [1], AutoML4ETC [2] and MLETC [3].

2. Background and Problem Definition

Encrypted traffic classification. Protocols such as TLS, QUIC and VPN tunnels hide the payload, so a classifier relies on what stays observable: packet sizes and directions, timing, handshake headers and statistical traces in the encrypted bytes. Formally, the task is to learn a function mapping a flow, usually its first few packets, to one label from a fixed set. Ciphers do not achieve perfect randomness in practice, which leaves exploitable structure [1].

Privacy issues. Classification is valuable for defence, but it also lets observers infer which applications a user runs without decrypting anything. The reviewed studies therefore mask IP addresses and ports, and AutoML4ETC also obfuscates the server name indication (SNI) and cipher fields, partly because Encrypted Client Hello will hide SNI [2], [3]. Masking also prevents models from memorising endpoints [1].

Limited labelled data. Correctly labelled encrypted traffic is scarce. Labels are often derived indirectly, for example by matching SNI values [2], and public datasets are small and imbalanced; [3] samples at most 500 flows per class. This motivates self-supervised pre-training on unlabelled traffic followed by fine-tuning [3], [4].

Real-time needs and drift. Classifiers must decide from the first few packets and run cheaply at line rate. AutoML4ETC reports that a production classifier degraded as traffic changed (data drift) and that retraining only partly restored it [2]. A practical classifier must therefore balance accuracy, size and speed, robustness over time, and privacy.

3. Summary of Selected Research Papers

3.1 Paper 1: ATVITSC (Liu et al., IEEE TIFS, 2024) [1]

Problem: Existing models extract spatial or temporal features inadequately, or lose temporal information when several packets are mixed into one image. The authors want both global relations among packets and local spatiotemporal patterns.

Method: Raw traffic is split into bidirectional sessions, and the first bytes of each packet's application data become a grayscale packet image; these are combined into a session image. Two parallel branches process it. The Packet Vision Transformer (PVT) uses packet, position and length embeddings with multi-head self-attention for global features. The Spatiotemporal Feature Extractor (STFE) applies residual attention convolution to each packet and a bidirectional LSTM across packets. A dynamic weighting layer with a temperature parameter fuses the two outputs and stops one branch from being ignored.

Dataset: USTC-TFC2016 (malicious traffic), ISCX Tor2016, ISCX VPN2016 and Cross-Platform (59 U.S. Android apps), using 16 packets of 256 bytes.

Results: Macro-F1 of 97.88% (VPN), 98.79% (Tor), 99.67% (malicious) and 94.90% (mobile). On Tor it beats the best baseline, TSCRNN, by 3.79 points. In ablation, PVT alone scores 95.85%, STFE alone 91.99%, both 98.00% and the full model 98.79%. Inference takes 32.54 ms per batch of 16, against 18.67 ms for FS-Net and 323.25 ms for HAN.

Contribution: Packet-level image construction, a global plus spatiotemporal dual-branch design, and temperature-controlled dynamic fusion.

3.2 Paper 2: AutoML4ETC (Malekghaini et al., IEEE TNSM, 2023) [2]

Problem: Hand-designed classifiers decay in production because of data drift, and manual re-tuning is slow and needs expertise. Operators also need light models that classify from the first packets.

Method: A neural architecture search (NAS) tool with a search space inspired by ENAS: one Normal and one Reduction cell of four nodes, each choosing inputs and operations (identity, separable convolution, pooling). Reinforcement learning, Monte Carlo tree search, evolutionary search and random search were compared, along with full training (40 epochs) and partial training (10 epochs), which saves about 75% of search time. Inputs are the headers of the first three TLS handshake packets or the single QUIC ClientHello.

Dataset: Six real datasets from the Orange mobile network (five TLS, one QUIC), at service and application level, plus public benchmarks including ISCXVPN2016 and UCDavis QUIC.

Results: On the May 2021 TLS data it reaches 82.86% accuracy with about 111.5 thousand parameters, against 77.55–78.75% with 12–22 million parameters for a hand-style CNN+MLP search space. It is about 1 to 60.1% more accurate than the baselines, uses over 50 times fewer parameters on average, and scores 100% on UCDavis QUIC. All four search algorithms reach top results within 0.25 points, so the search space matters more than the algorithm.

Contribution: A traffic-specific search space, regeneration of compact models for new data without expert tuning, accurate classification from a single QUIC packet, and open-source code.

3.3 Paper 3: MLETC (Park et al., IEEE Access, 2025) [3]

Problem: Pre-trained traffic models work at one level, bytes or packets, and ignore header fields, which are the units that carry protocol meaning.

Method: A DeBERTa model takes five packets of 100 bytes each (IP header, TCP/UDP header, 60 payload bytes). Two objectives are combined with equal weights: Masked Fields Prediction (MFP), which masks whole header fields and single payload bytes, and Same-Origin Flow Prediction (SOFP), which predicts whether packets 1–2 and 3–5 come from the same flow. Separate models are pre-trained for TCP and UDP and their predictions merged.

Dataset: About 30 GB of campus traffic for pre-training; ISCX VPN 2016, ISCX Tor 2016 and USTC-TFC 2016 for fine-tuning (500 flows per class, 7:3 split).

Results: On ISCX VPN, category accuracy is 98.75% (F1 0.9880) and application F1 is 0.9770; accuracy is about 99.5% on Tor and 99.0% on USTC-TFC. Without MFP accuracy drops to 90.48%, and without SOFP to 94.17%. It beats classical and plain deep models and is comparable to other pre-trained models such as ET-BERT [4].

Contribution: The first traffic pre-training scheme that models header fields as well as bytes and packets, and protocol-specific pre-trained models.

4. Comparative Analysis

4.1 Similarities

All three papers classify encrypted traffic from raw bytes of the first packets without hand-crafted features. They use deep networks with attention or convolution, residual connections, normalisation and dropout; ATVITSC and MLETC both rely on transformer attention. All truncate or pad early packets to a fixed size and mask addresses and ports. ISCX VPN 2016 appears in all three, ATVITSC and MLETC also share ISCX Tor and USTC-TFC, and Deep Packet, FS-Net and TSCRNN-style models serve as common baselines.

4.2 Differences

- **Focus.** ATVITSC improves feature extraction, AutoML4ETC automates model design and size, and MLETC improves self-supervised pre-training.
- **Input.** ATVITSC uses payload images of 16 packets, AutoML4ETC uses handshake headers of three TLS packets or one QUIC packet, and MLETC uses field-tokenised headers and payload of five packets.
- **Learning.** ATVITSC is supervised, AutoML4ETC is supervised with architecture search, and MLETC is self-supervised then fine-tuned.
- **Data and results.** Only AutoML4ETC uses multi-year real operator traffic, hence lower (82–96%) but more realistic accuracy, with models of about 0.1–0.3 million parameters. ATVITSC and MLETC report 98–99% on public benchmarks, and MLETC relies on a heavy DeBERTa model and two protocol models.

4.3 Strengths & Weaknesses

Paper	Strengths	Weaknesses
Paper 1: ATVITSC [1]	Packet-level images avoid information mixing; combines global and spatiotemporal features; stable dynamic fusion; strong on four datasets including mobile; reasonable speed.	Complex dual-branch model; controlled benchmark data; unproven on new encryption or dynamic data; hyperparameters tuned on one dataset.
Paper 2: AutoML4ETC [2]	Very small models; tested on real TLS and QUIC traffic over years; addresses drift; modular and open source.	Lower absolute accuracy; NAS is costly; labels approximated from SNI; no attention or transformer blocks yet.
Paper 3: MLETC [3]	Adds field-level information; learns from unlabelled data; both objectives shown to matter; high accuracy on three datasets.	Heavy model and two protocol models, inefficient for real time; ignores option fields and higher-layer protocols; compared with other papers' reported numbers, with minor text–table mismatches.

5. Discussion

The papers show three complementary routes to the same goal: better representations (ATVITSC), better learning signals (MLETC) and better architectures (AutoML4ETC). Their gains do not conflict, since a field-aware pre-trained model could seed an architecture search. There is a clear shift from CNNs and RNNs to attention-based and pre-trained models, as earlier in language and vision [3], a move to early classification from few packets [2], and growing reporting of model size and latency [1], [2].

Each paper is experimentally thorough, with ablations, hyperparameter studies and many baselines. However, benchmark accuracy of 98–99% leaves small differences that may come from preprocessing and sampling choices, and MLETC itself notes that comparisons are hard because settings differ [3]. The main gaps are old, small and controlled benchmark datasets, the drop to 82–95% on real operator traffic in [2], which suggests benchmarks overstate real performance, little attention to drift outside [2], almost no explanation of individual decisions, and few measurements on resource-limited devices.

6. Future Research Directions

- **Lightweight models.** Distillation, pruning and quantisation could bring pre-trained models to NAS-sized networks; [3] proposes DistilBERT-style models and [2] one-shot search.
- **Privacy.** Federated learning and differential privacy would let operators train a shared model without exchanging raw traffic.
- **Cross-domain and continual learning.** Models should transfer between networks and update incrementally as protocols change (TLS 1.3, QUIC, Encrypted Client Hello) [1], [2].
- **Explainable AI.** Attention visualisation and feature attribution should show which bytes or fields drive a decision, so operators can audit it.
- **Real-world evaluation.** Newer, larger datasets, evaluation over time, unknown-application handling and deployment-level latency tests are needed.

7. Conclusion

ATVITSC [1] improves feature extraction with packet images and a fused transformer and BiLSTM design, reaching macro-F1 of up to 99.67%. AutoML4ETC [2] automates design and yields models far smaller than hand-built ones on real TLS and QUIC traffic. MLETC [3] shows that field-aware pre-training reaches 98–99% accuracy on public data. Together they confirm a trend towards attention-based, pre-trained and automatically designed models and show that benchmark accuracy is close to saturation. The remaining challenges are efficiency, drift, privacy, explainability and realistic evaluation, which matter because encrypted traffic keeps growing and network security depends on understanding it without breaking encryption.

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